

- 3 (a) The green-eyed tree-frog, *Litoria genimaculata*, lives in tropical rainforests in Australia. Around 5 million years ago, the tree-frog population split into northern (**N**) and southern (**S**) populations that were separated by a dry area with no rainforest.

6500 years ago, the **N** and **S** tree-frog populations came into contact again as a warmer climate allowed rainforest to grow again in the area between the **N** and **S** populations.

Figure 3.1 is a map showing an area, labelled the contact zone, where **N** and **S** populations of tree-frog currently come into contact.

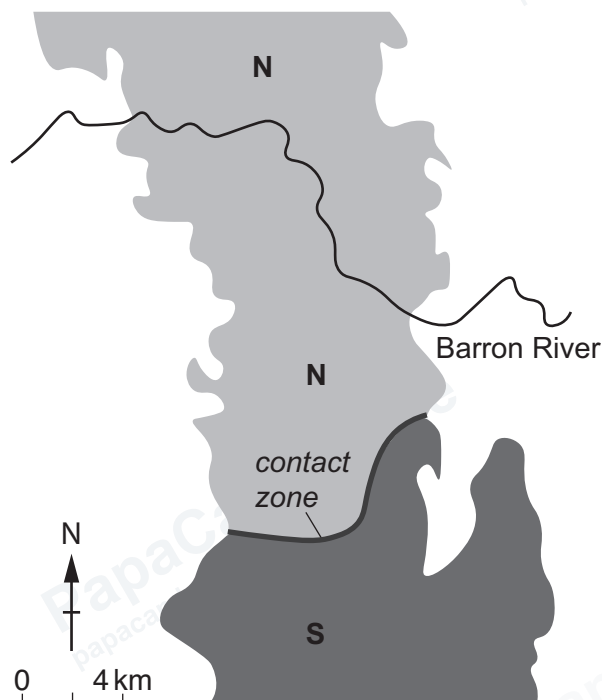


Figure 3.1

- (i) **N** tree-frogs and **S** tree-frogs can be distinguished by genetic techniques.

Outline the genetic techniques that could be used to distinguish between **N** and **S** tree-frog DNA.

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- (ii) Explain how a period of geographical separation, such as the one that occurred between the **N** and **S** tree-frog populations, can eventually lead to speciation.

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- (iii) Scientists were interested in tree-frogs in the contact zone.

The scientists made these observations:

- In the wild, **S** female tree-frogs are strongly attracted to the mating call of **S** male tree-frogs but are **not** attracted to the mating call of **N** male tree-frogs.
- In the laboratory, when an **S** female tree-frog is mated with a male **N** tree-frog, none of the fertilised eggs develop into offspring.

Suggest, with reasons, what these observations show about speciation between **N** tree-frogs and **S** tree-frogs.

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- (b) There is a large variation in body length within the population of **N** tree-frogs and within the population of **S** tree-frogs.

Describe the factors that interact to cause this variation in body length.

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